

2.3 代码编译器gcc

2.3.1 gcc简介

gcc是GNU Compiler Collection的简称，是GNU(GNU's Not Unix)开源项目的编译器套件。gcc 的初衷是为GNU操作系统专门编写的一款编译器，用于编译C代码。现如今已扩展为可以编译C++、Java、Objective-C等多种编程语言的集合。gcc本身也遵循GPL发行许可证，大名鼎鼎的linux就是基于gcc搭建的编译系统。gcc 官方网站 (<https://gcc.gnu.org/>)

GCC, the GNU Compiler Collection

The GNU Compiler Collection includes front ends for C, C++, Objective-C, Fortran, Ada, Go, D and Modula-2 as well as libraries for these languages (libstdc++ +...). GCC was originally written as the compiler for the GNU operating system. The GNU system was developed to be 100% free software, free in the sense that it respects the user's freedom.

We strive to provide regular, high quality releases, which we want to work well on a variety of native and cross targets (including GNU/Linux), and encourage everyone to contribute changes or help testing GCC. Our sources are readily and freely available via Git and weekly snapshots.

Major decisions about GCC are made by the steering committee, guided by the mission statement.



News

- [GCC 14.2 released](#) [2024-08-01]
- [GCC 11.5 released](#) [2024-07-19]
- [GCC 12.4 released](#) [2024-06-20]
- [GCC 13.3 released](#) [2024-05-21]
- [GNU Tools Cauldron 2024](#) [2024-05-07]
Prague, Czech Republic, September 14-16 2024
- [GCC 14.1 released](#) [2024-05-07]
- [GCC developer room at FOSDEM 2024: Call for Participation open](#) [2023-11-20]
FOSDEM 2024: Brussels, Belgium, February 3-4 2024
- [GNU Tools Cauldron 2023](#) [2023-09-05]
Cambridge, United Kingdom, September 22-24 2023
- [GCC 13.2 released](#) [2023-07-27]
- [GCC 10.5 released](#) [2023-07-07]
- [GCC Code of Conduct adopted](#) [2023-06-16]
- [GCC 11.4 released](#) [2023-05-29]
- [GCC 12.3 released](#) [2023-05-08]
- [GCC 13.1 released](#) [2023-04-26]
- [GCC BPF in Compiler Explorer](#) [2022-12-23]
Support for a nightly build of the bpf-unknown-none-gcc compiler has been contributed to Compiler Explorer (aka godbolt.org) by Marc Poulhiès
- [Modula-2 front end added](#) [2022-12-14]
The Modula-2 programming language front end has been added to GCC. This front end was contributed by Gaius Mulley.
- [GNU Tools Cauldron 2022](#) [2022-09-02]
Prague, Czech Republic and online, September 16-18 2022
- [GCC 13.2 released](#) [2023-08-10]

Supported Releases

GCC 14.2 (changes)
Status: 2024-07-11 (regression fixes & docs only).
Serious regressions. All regressions.

GCC 13.3 (changes)
Status: 2024-04-30 (regression fixes & docs only).
Serious regressions. All regressions.

GCC 12.4 (changes)
Status: 2024-06-20 (regression fixes & docs only).
Serious regressions. All regressions.

Development: GCC 15.0 (release criteria, changes)
Status: 2024-04-26 (general development).
Serious regressions. All regressions.

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